Contents

Index

0.1 Exersice

Exercise 0.1.1 ▸ Calculate

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove in two different ways that for every $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

Proof. 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \operatorname{Alt}(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \operatorname{Alt}(\beta \otimes \alpha)$$

Only need to show that

$$\operatorname{Alt}(\alpha \otimes \beta) = (-1)^{pq} (\beta \otimes \alpha)$$

Where

$$\operatorname{Alt}(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)\, \sigma (\alpha \otimes \beta)$$

Let $\tau = (p+1, \cdots, p+q, 1, 2, \cdots, p)$, then $\operatorname{sgn}(\tau) = (-1)^{pq}$. We have

$$\begin{aligned}
\operatorname{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)\, \sigma \circ \tau (\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)(\operatorname{sgn}\tau)(\sigma \circ \tau)(\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\operatorname{sgn}\sigma')\, \sigma' (\beta \otimes \alpha) \\
&= (-1)^{pq} \operatorname{Alt}(\beta \otimes \alpha)
\end{aligned}$$

2. Suppose

$$\alpha = \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} e^*_{i_1} \wedge \cdots \wedge e^*_{i_p}$$

$$\beta = \sum_{j_1 \cdots j_q} b_{j_1 \cdots j_q} e^*_{j_1} \wedge \cdots \wedge e^*_{j_q}$$

Then

$$\alpha \wedge \beta = \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} b_{j_1 \cdots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ}$$

$$\beta \wedge \alpha = \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta$$

$\square$

Exercise 0.1.2

Let V be a vector space of finite dimension and let V^* denotes the dual space. Prove that if $(\theta_1, \ldots, \theta_r)$ is a family of r covectors in V^* which are ***linearly dependent***, then the following wedge product:

$$\theta_1 \wedge \cdots \wedge \theta_r = 0.$$

Proof. Since $\theta_1 \wedge \cdots \theta_r$ is multi-linear and alternative, take any vector $v^1, \cdots, v^r$, we have

$$\theta_1 \wedge \cdots \wedge \theta_r \left(v^1, \cdots, v^r\right) = \det\left(\theta_i\left(v^j\right)\right)_{ij}$$

Since $\theta_1, \cdots, \theta_r$ are linealy dependent , the row vectors of the matrix $\left(\theta_i\left(v^j\right)\right)_{ij}$ are as well. Thus $\det\left(\theta_i\left(v^j\right)\right)_{ij} = 0$, and $\theta_1 \wedge \cdots \theta_r\left(v^1, \cdots, v^r\right) = 0$. Since $v^1, \cdots, v^r$ are arbitrary , $\theta_1 \wedge \cdots \wedge \theta_r = 0$. □

Exercise 0.1.3 ▶ Basis of Grassmann algebra

Let V be a vector space of finite dimension with basis $(e_1, \ldots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \ldots, e_n^*)$. Please prove that

$$(e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*)_{1 \le i_1 < \cdots < i_k \le n}$$

forms a basis of $\Lambda^k V^*$.

Proof. For $\alpha \in \Lambda^k V^*$, and $I = (i_1, \cdots, i_k)$. Define

$$\alpha_I := \alpha\left(e_{i_1}, \cdots, e_{i_k}\right)$$

If I has a reputed index , then $\alpha_I = 0$, and if $J = \sigma I$, $\alpha_J = (\operatorname{sgn}\sigma)\,\alpha_I$. We have

$$\sum_I{}' \alpha_I e_I^*\left(e_{j_1}, \cdots, e_{j_k}\right) = \sum_I{}' \alpha_I \delta_J^I = \alpha_J = \alpha\left(e_{j_1}, \cdots, e_{j_k}\right)$$

Thus $\alpha = \sum_I' \alpha_I e_I^*$. Which shows that $e_{i_1}^*, \cdots, e_{i_k}^*, 1 \le i_1 < \cdots < i_k \le n$ spands the $\Lambda^k V^*$. To show that they are linealy independent. Supppose that

$$\sum_I k_I e_I^* = 0$$

Both sides act on e_J, we have $k_J = 0$. □

Exercise 0.1.4 ▶ Grassmann algebra versus Polynomial functions on V.

Let V be a vector space of finite dimension and let V^* denotes the dual space. A p-multilinear map α on V is ***symmetric*** if

$$\alpha(v_1, \ldots, v_n) = \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(n)})$$

for all permutations σ. Such α defines a homogeneous polynomial $\tilde{\alpha}$

defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all $v \in V$. Then show that for symmetric multilinear maps α, β on V of respective p, q, the product $\alpha \circ \beta$ defined by

$$\alpha \circ \beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on V of degree $p+q$ and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v)\beta(v, \dots, v).$$

Could you compare with $\Lambda^{\bullet} V^*, \wedge$?

Exercise 0.1.5 ▸ Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere $S^n \subset \mathbb{R}^{n+1}$.
- The matrix groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$, these are Lie groups of matrice. Each time could you tell me what is the dimension of these groups.

Proof.

- Define F:

$$F\left(x^1, \cdots, x^{n+1}\right) = x_1^2 + \cdots + x_{n+1}^2$$

Then

$$S^n = F^{-1}(1)$$

$$\mathrm{d}F_x = \nabla F(x) = \left(2x_1, \cdots, 2_{x_{n+1}}\right) \neq 0, \forall x \in \mathbb{R}^{n+1}$$

By regular value theorem, $S^n \subseteq \mathbb{R}^{n+1}$ is a regular embeded submanifold of $\mathbb{R}^{n+1}$, which itself is a manifold.

- 1. $\mathrm{Mat}(n, \mathbb{R}) \simeq \mathbb{R}^2$ 是 n^2 维流形.
  2. $\mathrm{GL}(n, \mathbb{R}) = \{M \in \mathrm{Mat}(n, \mathbb{R}) : \det M \neq 0\}$ 作为 $\mathrm{Mat}(n, \mathbb{R})$ 的一个开子集也是一个 n^2 维流形.

3. $\mathrm{SL}(n,\mathbb{R}) = (M \in \mathrm{Mat}(n,\mathbb{R}))$: $\det M = 1$

$$\begin{aligned} \det : \mathrm{Mat}(n,\mathbb{R}) &\to \mathbb{R} \\ M &\mapsto \det M \end{aligned}$$

$SL(n,\mathbb{R}) = \det^{-1}(I)$. 取一个 $M \in \mathrm{Mat}(n,\mathbb{R})$, $\det M = 1$. 要证明 $(\mathrm{d}(\det))_M : \mathbb{R}^{n^2} \to \mathbb{R}$. 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}}(\det M) = A_{ij}(\text{代数余子式})$$

由于 $\det M = 1$. 至少有一个 $A_{ij} \neq 0$. 从而 $(\mathrm{d}(\det))_M$ 是满的.

4. 定义 $P : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})(\text{对称矩阵}) \simeq \mathbb{R}^{\frac{n(n+1)}{2}}$.

$$P(A) := A^\top A$$

则 $O(n,\mathbb{R}) = P^{-1}(I)$. 对于 $A \in O(n,\mathbb{R})$.

$$(\mathrm{d}P)_A : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})$$

与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于 P 本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A+\varepsilon B) - P(A) = \varepsilon(\mathrm{d}P)_A(B) + O(\cdots)$$

带入得到

$$\left(A^T + \varepsilon B^T\right)(A+\varepsilon B) - A^\top A = \varepsilon\left(B^\top A + A^\top B\right) + O(\varepsilon)$$

于是 $(\mathrm{d}P)_A(B) = B^\top A + A^\top B$. 只需要证明这个关于 B 的映射是满的. 为此, 解方程

$$A^\top B + B^\top A = C$$

取 $B = \frac{1}{2}AC$, 利用 A, C 对称, 带入得到

$$\frac{1}{2}A^\top AC + \frac{1}{2}C^\top A^\top A = \frac{1}{2}\left(C + C^\top\right) = C$$

故 $O(n,\mathbb{R})$ 是 $\mathrm{Mat}(n,\mathbb{R})$ 的一个 $\frac{n(n-1)}{2}$ 维正则子流形. 注意到对于 $A \in O(n,\mathbb{R})$, $\det A = 1$ 或 $\det A = -1$, 可以窥见事实上 $O(n,\mathbb{R})$ 有两个连通分支.

□

Exercise 0.1.6 ▸ Qunotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$ obtained by quotienting S^n by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is a smooth manifold.

Proof.

- 1. See $\mathbb{R}P^n$ is $S^2/\left(x \sim -x\right)$. The orentation changes at the equator. The map $A : S^2 \to S^2 . A\left(x\right) = -x$ induces an idendity map on $\mathbb{R}P^2$, $\mathrm{d}A$ send an orentation basis $\{e_1, e_2\}$ to one opposite to it .
  2. Use $\mathbb{R}P^2 = D^2/\{x \sim -x \text{ for } x \in \partial D^2 = S^1\}$. Then the neighborhood of the line at infinity is a thin ring near ∂D^2 with some quotient relation. The quotient is just the same at that of Möbius .

□

Exercise 0.1.7 ▸ Vector fields are derivation

Let M be a C^∞ manifold. Show that the left C^∞ module of derivation of the ring $C^\infty(M)$ i.e. linear maps $D : C^\infty(M) \to C^\infty(M)$ satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on M, smooth sections of TM.

Proof. Define

$$D\left(x\right) f = \left(Df\right)\left(x\right)$$

Then

$$D\left(x\right)\left(fg\right) =$$

□

1 Smooth Manifold

1.1 Topological Manifolds

Definition 1.1.1

Suppose M is a topological space. We say that M is a ***topological manifold of dimension n*** or a ***topological n-manifold*** if it has the following properties:

- M is a ***Hausdorff space***(T_2) : for every pair of distinct points $p, q \in M$, there are disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- M is ***second-countable***: there exists a countable basis for the topology of M.
- M is ***locally Euclidean of dimension n***: each point of M has a neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Remark.
The third property means, more specifically, that for each $p \in M$ we can find

- an open subset $U \subseteq M$ containing p,
- an open subset $\hat{U} \subseteq \mathbb{R}^n$, and
- a homeomorphism $\varphi : U \to \hat{U}$.

Remark.
A topological manifold M is locally compact, locally path-connected, seperable, Lindëlof, σ-finite, T_3, T_4, and paracompact.

1.1.1 Coordinate Chart

Definition 1.1.2 ▸ Coordinate Chart

Let M be a topological ***n-manifold***. A ***coordinate chart*** (or just a ***chart***) on M is a pair (U, φ), where U is an open subset of M and $\varphi : U \to \hat{U}$ is a homeomorphism from U to an open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Remark.

- By definition of a topological manifold, each point $p \in M$ is contained in the domain of some chart (U, φ).
- If $\varphi(p) = 0$, we say that the chart is ***centered at p***.
- If (U, φ) is any chart whose domain contains p, it is easy to obtain a new chart centered at p by subtracting the constant vector $\varphi(p)$.

1.2 Smooth Structures

Definition 1.2.1 ▸ Smoothly Compatible

Let M be a topological n-manifold.

- If $(U, \varphi), (V, \psi)$ are two charts such that $U \cap V \neq \varnothing$, the composite map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is called the ***transition map from φ to ψ***.
- Two charts (U, φ) and (V, ψ) are said to be ***smoothly compatible*** if either $U \cap V = \varnothing$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Remark.

- $\psi \circ \varphi^{-1}$ is a composition of homeomorphisms, and is therefore itself a homeomorphism.
- Since $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open subsets of $\mathbb{R}^n$, smoothness of this map is to be interpreted in the ordinary sense of having continuous partial derivatives of all orders.

Definition 1.2.2 ▸ Atlas

We define an ***atlas for M*** to be a collection of charts whose domains cover M. An atlas $\mathcal{A}$ is called a ***smooth atlas*** if any two charts in $\mathcal{A}$ are smoothly compatible with each other.

Definition 1.2.3 ▸ Maximal Atlas

A smooth atlas $\mathcal{A}$ on M is ***maximal*** if it is not properly contained in any larger smooth atlas. This just means that any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Definition 1.2.4 ▸ Smooth Structures and Smooth Manifolds

- If M is a topological manifold, a ***smooth structure on M*** is a maximal smooth atlas.
- A ***smooth manifold*** is a pair $(M, \mathcal{A})$, where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M.

2 Smooth Maps

2.1 Smooth Functions and Smooth Maps

2.1.1 Smooth Maps Between Manifolds

Definition 2.1.1 ▸ Smooth Maps Between Manifolds

Let M, N be smooth manifolds, and let $F : M \to N$ be any map. We say that F is a ***smooth map*** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

3 Sard's Theorem

3.1 Set of Measure Zero

Lemma 3.1.1

Suppose $A \subseteq \mathbb{R}^n$ is a compact subset whose intersection with $\{c\} \times \mathbb{R}^{n-1}$ has $(n-1)$- dimensional measure zero for every $c \in \mathbb{R}$. Then A has a n - dimensional measure zero .

Proof sketch.
把 A 的每一个“片” $A_c := \left\{(c,x) \in \{c\} \times \mathbb{R}^{n-1} : (c,x) \in A\right\}$ 的开覆盖, 缩短成里面的一个开的“方片”. 这些“方片”构成 A 的一个开覆盖, 取出一个有限子覆盖计算其 Volume, 将“方片”的相交部分抹去, 使得总体积可控.

Proof. Take $[a,b] \subseteq \mathbb{R}$ such that $A \subseteq [a,b] \times \mathbb{R}^{n-1}$.

Let $A_c := \left\{(c,x) \in \{c\} \times \mathbb{R}^{n-1} : (c,x) \in A\right\}$. The projection from A to $[a,b] \times R^{n-1}$ is continious map from a compact set to a Hausdorff space, thus is a proper map . It follows that A_c is a compact subspace of A for all $c \in [a,b]$.

Take any $\delta > 0$. There exists a finite open cover $C_1, \cdots, C_k$ of A_c , such that the total volumn is less than δ . Denote the open subset $C_1 \cup \cdots \cup C_k \subseteq A_c$ by U_c . We claim that there is a open interval J_c containing c and contained in U_c , such that A_c is contained in the intersection of A with $J_c \times \mathbb{R}^{n-1}$: otherwise, there is a suquence $\{(c_i, x_i)\}$, such that $c_i \to c$, $x_i \notin U_c$; but is passing to a subsequence $x_i \to A_c \setminus U_c$, which is a contradiction to $A_c \subseteq U_c$.

The intervals $\{J_c : c \in [a,b]\}$ forms a open cover of $[a,b]$, so there is a finitely many $c_1 < c_2 < \cdots < c_m$ such that $J_{c_1}, \cdots, J_{c_m}$ forms a finite open cover of A . We may assume that the total length is less than $2\,|b-a|$ by shrinking the intersection of the members. Then $J_{c_1}, \cdots, J_{c_m}$ is a open cover of A with total volumn less than $2\delta\,|b-a|$, which can be arbitrally small . Hence A has a n- dimensional measure zero . □

Proposition 3.1.2

Suppose A is an open or closed subset of $\mathbb{R}^{n-1}$ or $\mathbb{H}^{n-1}$, and $f : A \to \mathbb{R}$ is a continuous function. Then the graph of f has measure zero in $\mathbb{R}^n$.

Proof sketch.
对于 A 是紧集的情况，令 $P^n = \{\text{Sets are n-dimensional measure zero}\}$. 上面的引理给出了 $\forall c, P_c^{n-1} \implies P^n$ 的结论. 因此降维操作是无损的，从而归纳法有效.

Proof. First assume that A is compact. We prove the theorem in this case by induction on n. When $n = 1$, it is trivial because the graph of f is at most a point. To prove the inductinve step, we appeal to lemma above. For each $c \in \mathbb{R}$, the intersection of the graph of f with $\{c\} \times \mathbb{R}^{n-1}$ is just the graph of the restriction of f to $\left\{x \in A : x^1 = c\right\}$, which is in turn the graph of continuous functions of $(n-2)$ variables. It follows by induction that each such graph has $(n-1)$-dimensional measure zero, and thus by lemma above, the graph of f itself has n-dimensional measure zero.

If A is noncompact, it is a countable union of compact subsets, so the graph of f is a countable union of sets of measure zero. □

Corollary 3.1.3

Every proper affine subspace of $\mathbb{R}^n$ has measure zero in $\mathbb{R}^n$.

Proposition 3.1.4 ▶ Diffeomorphism-invariant of Measure Zero

Supposer $A \subseteq \mathbb{R}^n$ has measure zero and $F : A \to \mathbb{R}^n$ is a smooth map. Then $F(A)$ has measure zero.

Definition 3.1.5

If M is a smooth n- manifold with or without boundary, we say that a subset $A \subseteq M$ has ***measure zero in*** M if for every smooth chat (U, φ) for M, the subset $\varphi(A \cap U) \subseteq \mathbb{R}^n$ has n- dimensional measure zero.

Lemma 3.1.6

Let M be a smooth n-manifold with or without boundary and $A \subseteq M$. Suppose that for some collection $\{(U_\alpha, \varphi_\alpha)\}$ of smooth charts whose domains cover A, $\varphi_\alpha(A \cap U_\alpha)$ has measure zero in $\mathbb{R}^n$ for each α . Then A has measure zero in M .

Proof. Let (V, ψ) be any smooth chat for M . We need to show that $\psi(A \cap V)$ has measure zero . There exists countably many U_α's covering $A \cap V$. We write

$$\psi(A \cap V \cap U_\alpha) = \left(\psi \circ \varphi_\alpha^{-1}\right) \circ \varphi_\alpha(A \cap V \cap U_\alpha)$$

$\varphi_\alpha(A \cap V \cap U_\alpha)$ is a subset of measure zero set $\varphi_\alpha(A \cap U_\alpha)$, thus has measure zero as well. It follows from Proposition 2.1.4 that $\psi(A \cap V \cap U_\alpha)$ has measure zero . Then $\psi(A \cap V)$ is a countable unions of measure zero sets , thus is measure zero as well. □

Exercise 3.1.7

Let M be a smooth manifold with or without boundary. Show that a countable union of sets of measure zero in M has measure zero.

Proof. Suppose $A_1, A_2, \cdots$ are sets of measure zero in M . Let $A = \bigcup_{i=1}^{\infty} A_i$. Take charts $\left(U_{ij}, \varphi_{ij}\right), j = 1, 2, \cdots$ such that $A_i \subseteq \bigcup_{j=1}^{\infty} U_{ij}$. Then $\left\{U_{ij}\right\}_{i,j}^{\infty}$ are countably many open sets that containing A , such that for each i, j ,

$$\varphi_{ij}\left(A \cap U_{ij}\right) \subseteq \bigcup_{i=1}^{\infty} \varphi_{ij}\left(A_i \cap U_{ij}\right)$$

is a countable union of measure zero sets, thus itself is measure zero. Then it follows from Lemma 2.1.6 that A is of measure zero. □

Proposition 3.1.8

Suppoes M is a smooth manifold with or without boundary and $A \subseteq M$ has measure zero in M. Then $M \setminus A$ is dense in M.

Proof. If $M \setminus A$ is not dense. Then A contains a nonempty open subset of M, which implies that there is a chat (V, ψ) , such that $\psi(A \cap V)$ is a nonemtpy open subsets of $\mathbb{R}^n$. It is imposible that $\psi(A \cap V)$ has measure zero at the same time that it is open. □

4

Integral Curves and Flows

4.1 Integral Curves

Definition 4.1.1 ▸ Integral Curve

If V is a vector field on M, an ***integral curve of*** V is a differentiable curve $\gamma : J \to M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the ***starting point of*** γ.

Finding integral curves boils down to sloving a system of ordinary differential equations in a smooth chat. Suppose V is a smooth vector field on M and $\gamma : J \to M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = \left(\gamma^1(t), \cdots, \gamma^n(t)\right)$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)} = V^i(\gamma(t)) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equatoins:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \ldots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \ldots, \gamma^n(t)). \end{aligned}$$

Proposition 4.1.2

Let V be a smooth vector field on a smooth manifold M . For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p .

Proof. The ODE system corosponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. The the proposition follows from the Picard-Lindelöf Theorem. $\square$

4.2 Exercise